

Research interests

My current research interests lie primarily in the area of **post-quantum cryptography**, whose aim is to study cryptographic systems that are considered secure even against quantum algorithms. In particular, my research focuses on the **cryptanalysis** of schemes coming from **code-based cryptography** and **multivariate cryptography** through techniques borrowed from **computational algebra**, such as **Gröbner bases**. I am also interested in several aspects of **algebraic coding theory**.

Work Experience

Associate professor

Université de Montpellier, LIRMM

- Teaching at the Bachelor and Master in Computer Science of the University of Montpellier
- Research within the ECO team at LIRMM

Montpellier, France

since September 2025

Postdoctoral researcher

CISPA - Helmholtz Center for Information Security

- Algorithmic Cryptology group led by Antoine Joux

Sankt Ingbert, Germany

November 2023 - August 2025

Research Engineer

Inria Paris Centre

- Project-team COSMIQ led by Jean-Pierre TILICH

Paris, France

April 2023 - October 2023

Education

Ph.D. in Computer Science

Inria Paris Centre and Sorbonne University

- **Research interests:** Post-quantum cryptography, Code-based Cryptography, Algebraic coding theory, Gröbner bases, Algebraic cryptanalysis
- **Thesis title:** Algebraic techniques for decoding Reed-Solomon codes and cryptanalyzing McEliece-like cryptosystems
- **Thesis advisor:** Jean-Pierre TILICH
- **Defence date:** April 7th, 2023

Paris, France

October 2019 - March 2023

Master in Mathematics, Curriculum “Coding Theory and Cryptography”

University of Trento

- **Final Mark:** 110/110 cum laude (full marks with honors)
- **Thesis title:** Efficient decoding algorithms for QC-LDPC and QC-MDPC code-based cryptosystems
- **Supervisors:** Prof. Marco BALDI, Prof. Massimiliano SALA
- **Defence date:** July 17th, 2019

Trento, Italy

October 2017 - July 2019

Bachelor in Mathematics

University of Parma

- **Final Mark:** 110/110 cum laude (full marks with honors)
- **Thesis title:** Lattice-based cryptography
- **Supervisor:** Prof. Alessandro ZACCAGNINI
- **Defence date:** October 24th, 2017

Parma, Italy

October 2014 - October 2017

Diploma in Piano

Conservatory of Music of Parma

- **Description:** Academic diploma equivalent to a Bachelor degree

Parma, Italy

October 2008 - September 2017

Maturity diploma

Scientific High School G. Marconi, Parma

Parma, Italy

September 2009 - July 2014

Achievements and Prizes

- 2024 **TII McEliece Challenges**, Prize of 10000\$ for winning the *Theoretical Key-Recovery Algorithms* track with the coauthored article “A New Approach Based on Quadratic Forms to Attack the McEliece Cryptosystem”
- 2023 **ERCIM “Alain Bensoussan” Postdoctoral Fellowship**, (refused)
- 2014 **Indam Scholarship**, Merit-based scholarship for students starting a Bachelor in Mathematics in Italy (40 scholarships in total, classified 15th in Italy)
- 2014 **Bronze Medal**, Italian Mathematical Olympiads
- 2013 **Bronze Medal**, Italian Mathematical Olympiads

Computer/Programming Skills

MAGMA, SAGEMATH, C, C++, PYTHON, JAVA, MATLAB, R, $\text{\LaTeX}$, COQ

Languages

- English** Full professional proficiency
- Italian** Native language
- French** Full professional proficiency